

## Parallel Heat Transfer and Safe Insulation Design Worksheet

Name(s):

Date:

### Educational Purpose

This digital experiment examines heat loss from a hot cylindrical vessel through a steel wall and insulation, followed by convection and radiation from the outer surface.

### Student Learning Objectives

1. Identify the heat-flow path from the process fluid to the ambient and distinguish series from parallel mechanisms.
2. Explain how convection coefficient and surface emissivity affect the two outer-surface heat-loss paths.
3. Explain how insulation thickness changes thermal resistance, total heat loss, and outer-surface temperature.
4. Use simulation data and governing equations to calculate heat rates, thermal resistances, and temperature changes.
5. Determine the minimum selectable insulation thickness that satisfies the outer-surface safety limit.

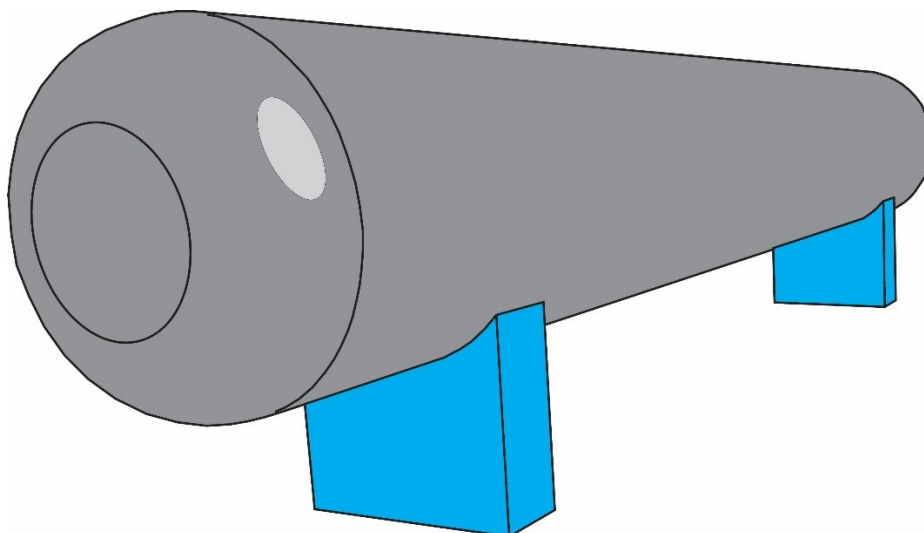
### Before Starting the Digital Experiment: Whole-System Prediction

This reaction vessel must maintain a constant internal temperature. We want to apply insulation around the steel vessel to reduce heat loss and keep the outer surface safe for people nearby.

Model assumptions: the inner steel wall is fixed at 800 °C; inside film resistance is neglected. Insulation conductivity is constant at 0.080 W/(m·K). Air and radiative surroundings are both at 25 °C.

How do you think heat will travel from the hot reactor interior to the surrounding air?

*Vessel Properties:*  $D_i = 1.00 \text{ m}$ ; *steel thickness* = 0.025 m;  $L = 6.00 \text{ m}$ ;  $k_{\text{steel}} = 45 \frac{\text{W}}{\text{m}\cdot\text{K}}$ ;



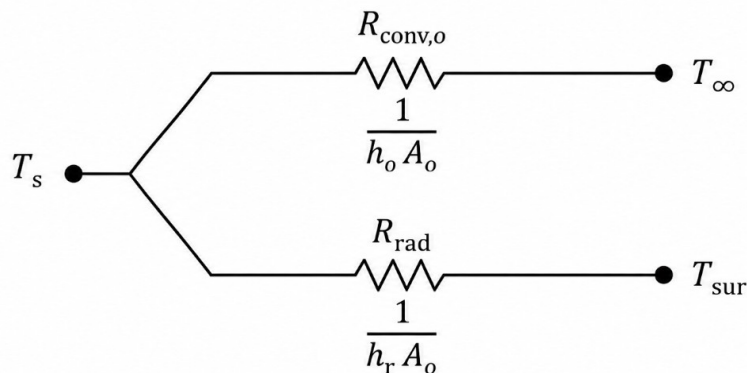
$$T_{\text{Air},\infty} = 25 \text{ }^{\circ}\text{C}$$

Predict how increasing insulation thickness will change total heat loss and outer-surface temperature. Explain why.

Predict how a larger convection coefficient or a smaller emissivity would change the rate of heat loss.

### Experiment 1: Parallel Convection and Radiation heat transfer (Case A, Trials 1–3)

Purpose: isolate each heat transfer path to the surroundings, then activate both paths together.



THERMAL-NETWORK FOCUS FOR 1–3: CONVECTION AND RADIATION BRANCH FROM THE SAME OUTER-SURFACE NODE AND THEREFORE ACT IN PARALLEL.

NOTE: CONVECTION TRANSFERS HEAT TO THE SURROUNDING AIR, WHILE RADIATION TRANSFERS HEAT TO THE SURROUNDING SURFACES  
The zero values of  $\epsilon$  and  $h$  are idealized limiting cases.

#### Procedure:

1. Click “Parallel paths”. Fixed settings:  $T_{\text{in}} = 800^\circ\text{C}$ , insulation thickness  $t_{\text{ins}} = 0.100\text{ m}$ ,  $k_{\text{ins}} = 0.080\text{ W/(m}\cdot\text{K)}$ .
2. For Trial 1, set  $\epsilon = 0.00$  and  $h = 8\text{ W/(m}^2\cdot\text{K)}$ . Click “Run and record trial” and copy the outputs.
3. For Trial 2, change  $\epsilon$  to 0.85 and  $h$  to  $0\text{ W/(m}^2\cdot\text{K)}$ . Click “Run and record trial” and copy the outputs.
4. For Trial 3, use  $\epsilon = 0.85$  and  $h = 8\text{ W/(m}^2\cdot\text{K)}$ . Click “Run and record trial” and copy the outputs.

**Table 1.** Experimental data for parallel surface paths

| Trials | $\varepsilon$ | $h[\text{W/m}^2\cdot\text{K}]$ | $T_s[^\circ\text{C}]$ | $\dot{Q}_{\text{conv}}[\text{kW}]$ | $\dot{Q}_{\text{rad}}[\text{kW}]$ | $\dot{Q}_{\text{total}}[\text{kW}]$ |
|--------|---------------|--------------------------------|-----------------------|------------------------------------|-----------------------------------|-------------------------------------|
| 1      | 0.00          | 8                              |                       |                                    |                                   |                                     |
| 2      | 0.85          | 0                              |                       |                                    |                                   |                                     |
| 3      | 0.85          | 8                              |                       |                                    |                                   |                                     |

**Questions After Experiment 1**

Use the relationships below. Here,  $h$  is the convection coefficient,  $\varepsilon$  is emissivity,  $A_o$  is the insulated outside area,  $T_s$  is the outer-surface temperature, and  $T_\infty$  is the ambient temperature. Use kelvin in the radiation equation.

$$\dot{Q}_{\text{conv}} = h A_o (T_s - T_\infty)$$

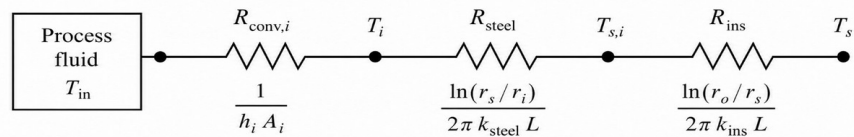
$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A_o (T_{s,K}^4 - T_{\infty,K}^4)$$

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$

- Which run isolates convection and which run isolates radiation?
- In trial 3, which of the parallel heat transfer contributes more to the heat loss?
- Why can the trial 1 and 2 heat rates not simply be added to predict trial 3?
- Why do you see a shiny surface (smaller emissivity) on the process vessels?

**Experiment 2: Series Conduction Through Steel and Insulation (Case A, Trials 4–6)**

Purpose: Observe how adding insulation resistance changes heat loss and outer-surface temperature.



THERMAL-NETWORK FOCUS FOR TRIALS 4–6: PROCESS → STEEL WALL → INSULATION → OUTER SURFACE IS A SERIES CONDUCTION PATH. The inside-film resistance shown in the schematic is neglected in this simulation.

**Procedure**

1. Click “Continue to series conduction”. Fixed settings:  $T_{in} = 800\text{ }^{\circ}\text{C}$ ,  $k_{ins} = 0.080\text{ W/(m}\cdot\text{K)}$ ,  $h = 8\text{ W/(m}^2\cdot\text{K)}$ ,  $\varepsilon = 0.85$ .
2. Run trial 4 with  $t = 0.000\text{ m}$  and record all outputs.
3. Run trial 5 with  $t = 0.100\text{ m}$ . Change only the insulation thickness and record all outputs.
4. Run trial 6 with  $t = 0.150\text{ m}$ . Again, change only the insulation thickness and record all outputs.
5. Check that trials 4–6 use identical values for every input except insulation thickness.

**Table 2.** Experimental data for insulation thickness

| <i>Trials</i> | $t_{ins}[\text{m}]$ | $T_s[{}^{\circ}\text{C}]$ | $\dot{Q}_{cond}[\text{kW}]$ | $\dot{Q}_{conv}[\text{kW}]$ | $\dot{Q}_{rad}[\text{kW}]$ | $\dot{Q}_{total}[\text{kW}]$ |
|---------------|---------------------|---------------------------|-----------------------------|-----------------------------|----------------------------|------------------------------|
| 4             | 0.000               |                           |                             |                             |                            |                              |
| 5             | 0.100               |                           |                             |                             |                            |                              |
| 6             | 0.150               |                           |                             |                             |                            |                              |

## Questions After Experiment 2

Use the relationships below. The radii  $r_i$ ,  $r_s$ , and  $r_o$  are the vessel inner radius, bare-steel outer radius, and insulated outer radius;  $L$  is the vessel length;  $k$  is the thermal conductivity; and  $R$  is the cylindrical thermal resistance.

$$R_{steel} = \frac{\ln(r_s / r_i)}{2 \pi L k_{steel}}$$

$$R_{ins} = \frac{\ln(r_o/r_s)}{2\pi L k_{ins}}$$

$$\dot{Q}_{cond} = \frac{T_{in} - T_s}{R_{steel} + R_{ins}}$$

- Describe the changes in  $T_s$  and  $\dot{Q}_{total}$  from trial 4 to 5 to 6.
- Why are the steel-wall and insulation resistances in series rather than in parallel?
- Does the additional 0.050 m of insulation from trial 5 to 6 appear to have the same effect as the first 0.100 m from trial 4 to 5?

### Experiment 3: Outer-Surface Safety Limit (Case B, Trials 7–8)

Purpose: find the minimum selectable insulation thickness to keep the touchable outer surface at or below 60 °C.

ASTM International. (2020). *ASTM C1055-20: Standard guide for heated system surface conditions that produce contact burn injuries*. ASTM International. <https://doi.org/10.1520/C1055-20>

## Procedure

**Click “Continue to safety limit”**

1. Confirm the fixed settings:  $T_{in} = 800\text{ }^{\circ}\text{C}$ ,  $k_{ins} = 0.080\text{ W/(m}\cdot\text{K)}$ ,  $h = 8\text{ W/(m}^2\cdot\text{K)}$ ,  $\varepsilon = 0.85$ .
2. Use insulation-thickness increments of  $0.005\text{ m}$ . Increase  $t$  until the first setting with  $T_s \leq 60\text{ }^{\circ}\text{C}$  is found.
3. Click “Record current trial” to save that first passing setting as trial 7, including both surface-path heat rates.
4. Reduce  $t_{ins}$  by exactly  $0.005\text{ m}$ ; the display updates automatically. Click “Record current trial” to save this immediately thinner setting as trial 8.
5. Mark each run Pass or Fail by comparing the recorded outer-surface temperature with  $60\text{ }^{\circ}\text{C}$ .

**Table 3.** Experimental data for the surface-safety search

| <i>Trials</i>          | $t_{ins}[m]$ | $T_s[^\circ C]$ | $\dot{Q}_{conv}[kW]$ | $\dot{Q}_{rad}[kW]$ | $\dot{Q}_{total}$<br>[kW] | <i>Pass / Fail</i> |
|------------------------|--------------|-----------------|----------------------|---------------------|---------------------------|--------------------|
| 7<br>first pass        |              |                 |                      |                     |                           |                    |
| 8<br>one step<br>lower |              |                 |                      |                     |                           |                    |

### Questions After Experiment 3

*outer-surface safety limit = 60 °C.*

Use  $T_s \leq 60^\circ C$  for the safety decision.  $T_s$  is the touchable outer-surface temperature

$$T_s \leq 60^\circ C \text{ and } t_{ins8} = t_{ins7} - 0.005 m$$

7. Why is the outer-surface temperature, rather than the process temperature, used for the touch-safety decision?

8. Why must the setting immediately below trial 7 be tested before trial 7 can be called the minimum selectable safe thickness?

9. At trial 7, which outer-surface mechanism contributes more heat loss? How might that affect the safe thickness if  $h$  or  $\epsilon$  changed?

### Homework Problems

Use the experimental results recorded in Tables 1–3. Show the equation, substitution, units, and final answer for each calculation. Use watts in thermal-resistance calculations and kelvin in the radiation equation.

#### Homework 1: Convection and Radiation in Parallel

**Question 1.** Calculate the isolated convection and radiation heat losses.

## Worksheet: Parallel Heat Transfer and Safe Insulation Design

For Trials 1–3, use the fixed insulation thickness of 0.100 m. First calculate the insulated outer radius and outer-surface area:

$$r_o = r_s + t_{ins} \text{ and } A_o = 2\pi r_o L$$

Using the outer-surface temperature recorded for Trial 1, calculate the convection heat loss:

$$\dot{Q}_{conv, trial1} = h A_o (T_{s, trial1} - T_\infty)$$

Using the outer-surface temperature recorded for Trial 2, calculate the radiation heat loss:

$$\dot{Q}_{rad, trial2} = \varepsilon \sigma A_o (T_{s, trial2, K}^4 - T_{\infty, K}^4)$$

Report both heat rates in kilowatts. Compare each calculated value with the corresponding output in Table 1 and state whether it agrees within normal rounding.

### Homework 2: Complete Thermal Network and Insulation Performance

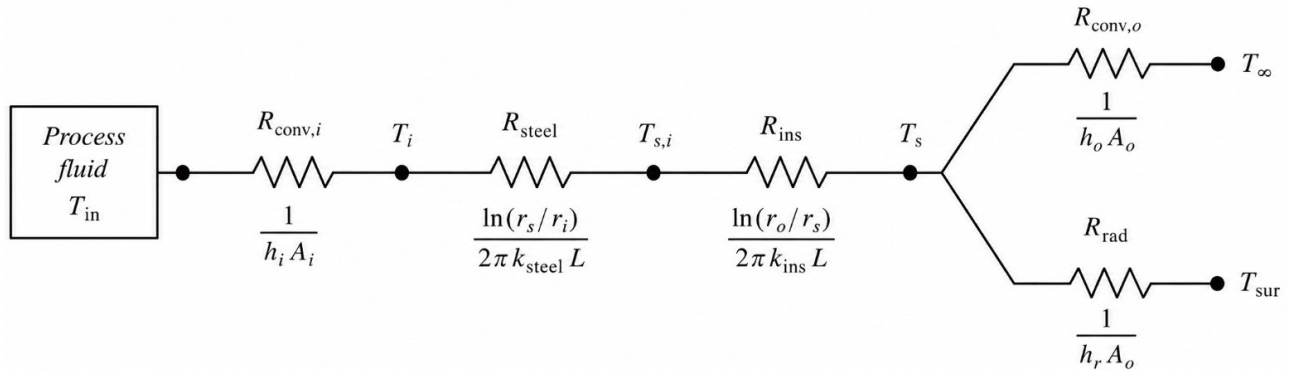


FIGURE 1. THERMAL NETWORK FOR THE COMPLETE VESSEL SYSTEM. The inside-film resistance is neglected; the specified temperature is applied at the inner steel wall.

**Question 2. Analyze the complete vessel system shown in Figure 1.** Use the Trial 3 results. Heat passes through the steel wall and insulation in series and then leaves the outer surface by convection and radiation in parallel.

1. Use the insulated outer radius calculated in Question 1. Calculate the steel-wall and insulation resistances:

$$R_{steel} = \frac{\ln(r_s / r_i)}{2\pi L k_{steel}}$$
$$R_{ins, trial3} = \frac{\ln(r_{o, trial3} / r_s)}{2\pi L k_{ins}}$$

2. Calculate the total heat-transfer rate leaving the outer surface:

$$\dot{Q}_{total, trial3} = \dot{Q}_{conv, trial3} + \dot{Q}_{rad, trial3}$$

3. Convert the total heat-transfer rate from kilowatts to watts. Then calculate the steel–insulation interface temperature and the predicted outer-surface temperature:

$$T_{interface} = T_{in} - \dot{Q}_{total, trial3} R_{steel}$$
$$T_{s, predicted} = T_{interface} - \dot{Q}_{total, trial3} R_{ins, trial3}$$

4. Compare the predicted outer-surface temperature with the Trial 3 simulation result. In one or two sentences, explain why the heat rate through the steel and insulation must equal the combined convection and radiation heat rates at steady state.

### Homework 3: Outer-Surface Safety

**Question 4. Confirm the minimum selectable safe thickness.** Use the surface-safety requirement and the required relationship between Trials 7 and 8:

$$T_s \leq 60^\circ C \text{ and } t_{ins, trial8} = t_{ins, trial7} - 0.005 m$$

For Trials 7 and 8, report the insulation thickness and outer-surface temperature, classify each trial as Pass or Fail, and confirm that the Trial 8 thickness is exactly 0.005 m below the Trial 7 thickness. Explain briefly why these results show that Trial 7 is the minimum safe thickness selectable using 0.005 m increments.